AI ACADEMY · LEVEL 3 · GRADE 8 · WEEK 26 · STUDENT LESSON

Evaluation Sets

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

Before you start

AI-2 Week 11: development and final evaluation need separate cases.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Explain the mechanism and terminology of system evaluation sets. Try the recall before opening notes.

Your learning target

I can explain the **mechanism** and terminology of system **evaluation set**s.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Quality Rubrics. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Block to Python Bridge. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Conversation State and Memory. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 26: **Evaluation Set**s

A good eval set represents real use, edge cases, and known risks.



CREATOR MISSION

Author ten tests with **expected behavior**.

Creator name and date:

> **Write or draw your reasoning and evidence.**

Readiness for the connected explanation

Recall Latency cost and resource use. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

Readiness for Ranking and retrieval evaluation

Recall Lexical and vector retrieval. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

> **My readiness explanation**

This week’s end goal

By the end, I can explain **evaluation set**s using a traced example and a stated limitation.

Notice the evidence

> Investigate the supplied Evaluation Sets case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



Start with this puzzle

MYSTERY

The learners must use **evaluation set**s to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

Write or draw your reasoning and evidence.

My **prediction** and the clue supporting it:

Write or draw your reasoning and evidence.

A closer explanation

A system **evaluation set** contains cases with expected behaviour and scoring rules. It should cover retrieval, support, fallback and permissions rather than only easy answerable questions.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Apply the connected idea: Ranking and retrieval evaluation

A ranking needs task-specific relevance judgments and a declared **cutoff**. Precision and recall have different denominators.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

STUDENT LESSON · 3 OF 18

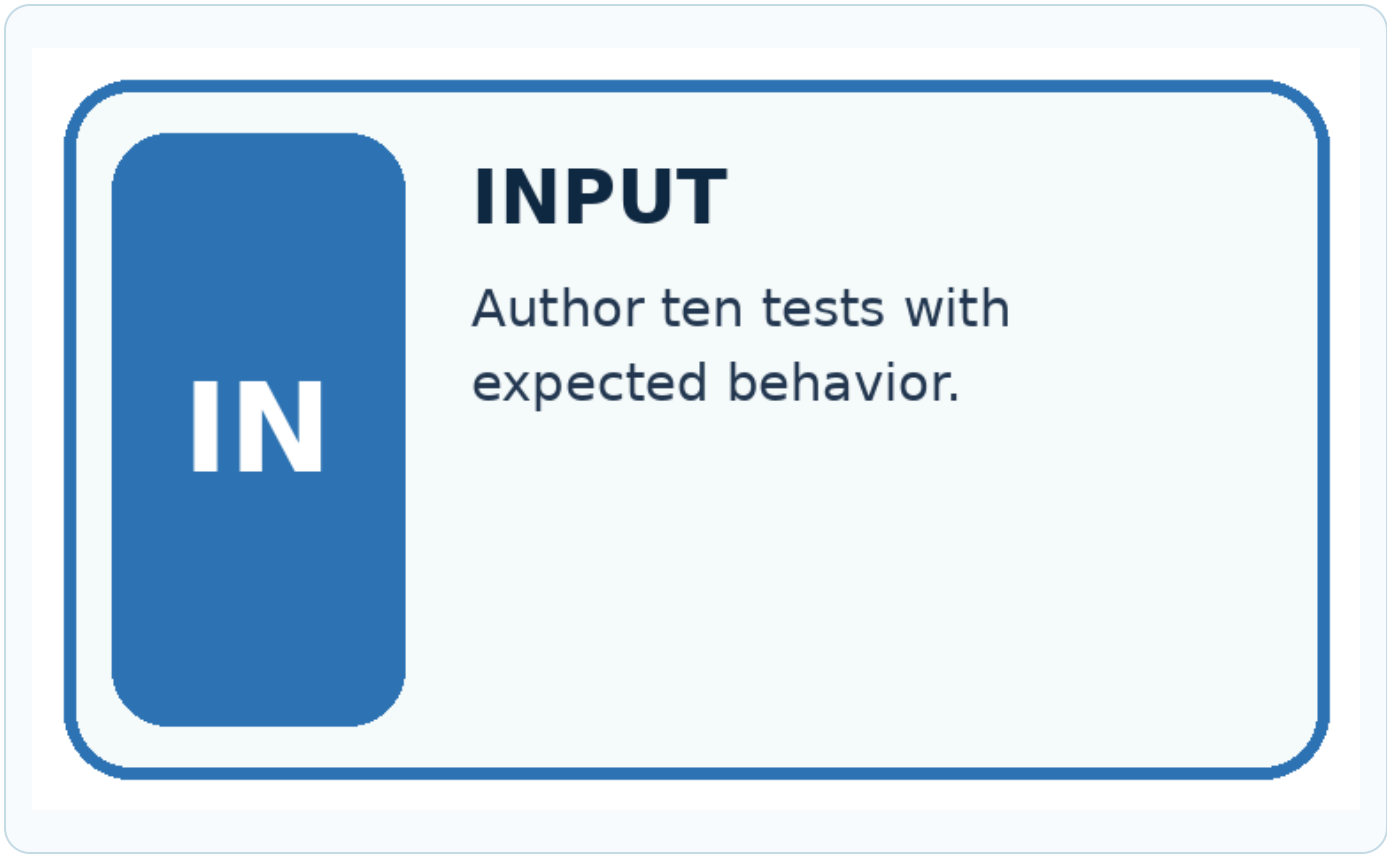
See how it works

> Follow the mechanism for Evaluation Sets. Connect each labelled input, deciding step and result.

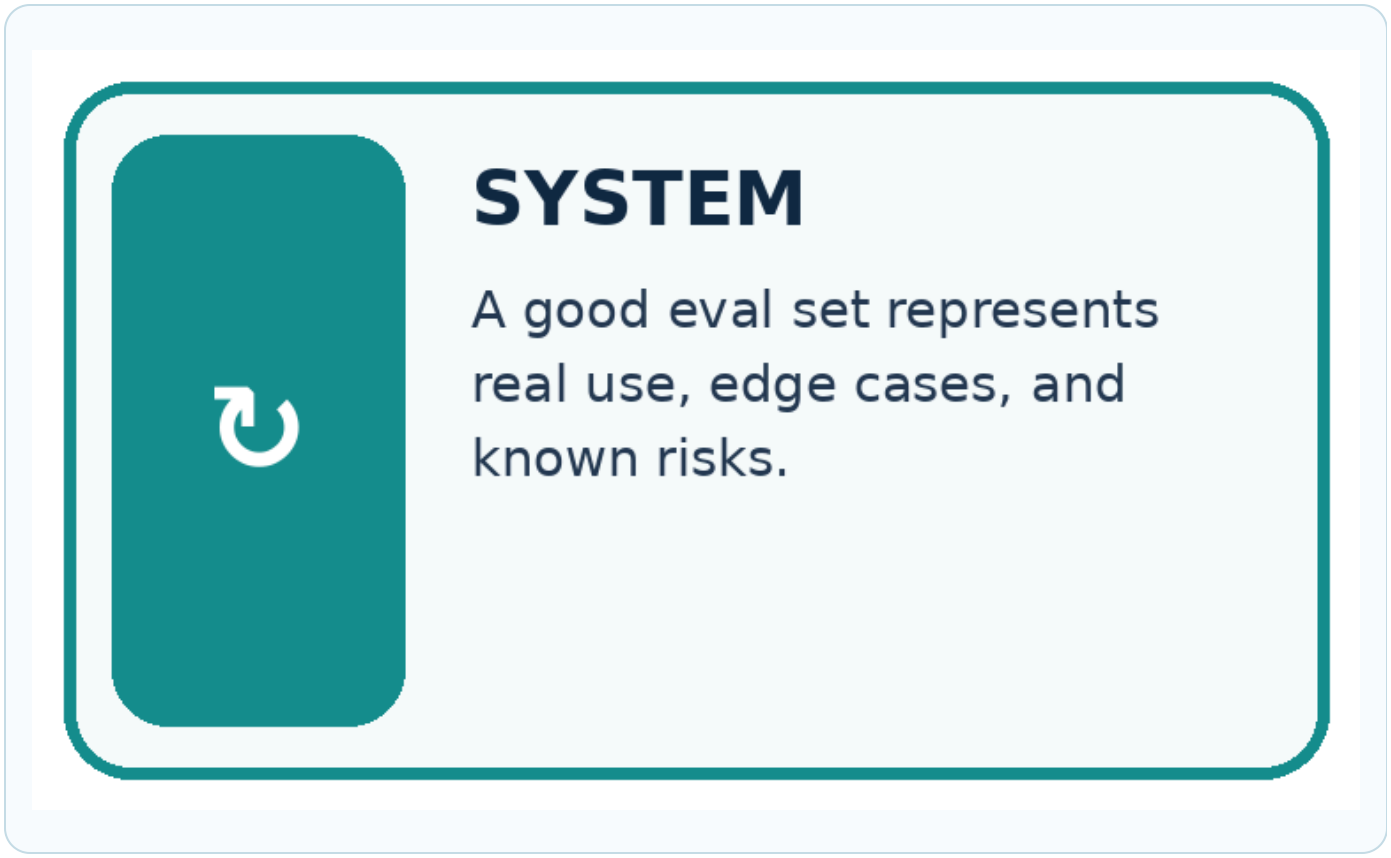
VISUAL MECHANISM

Input → Process → Output

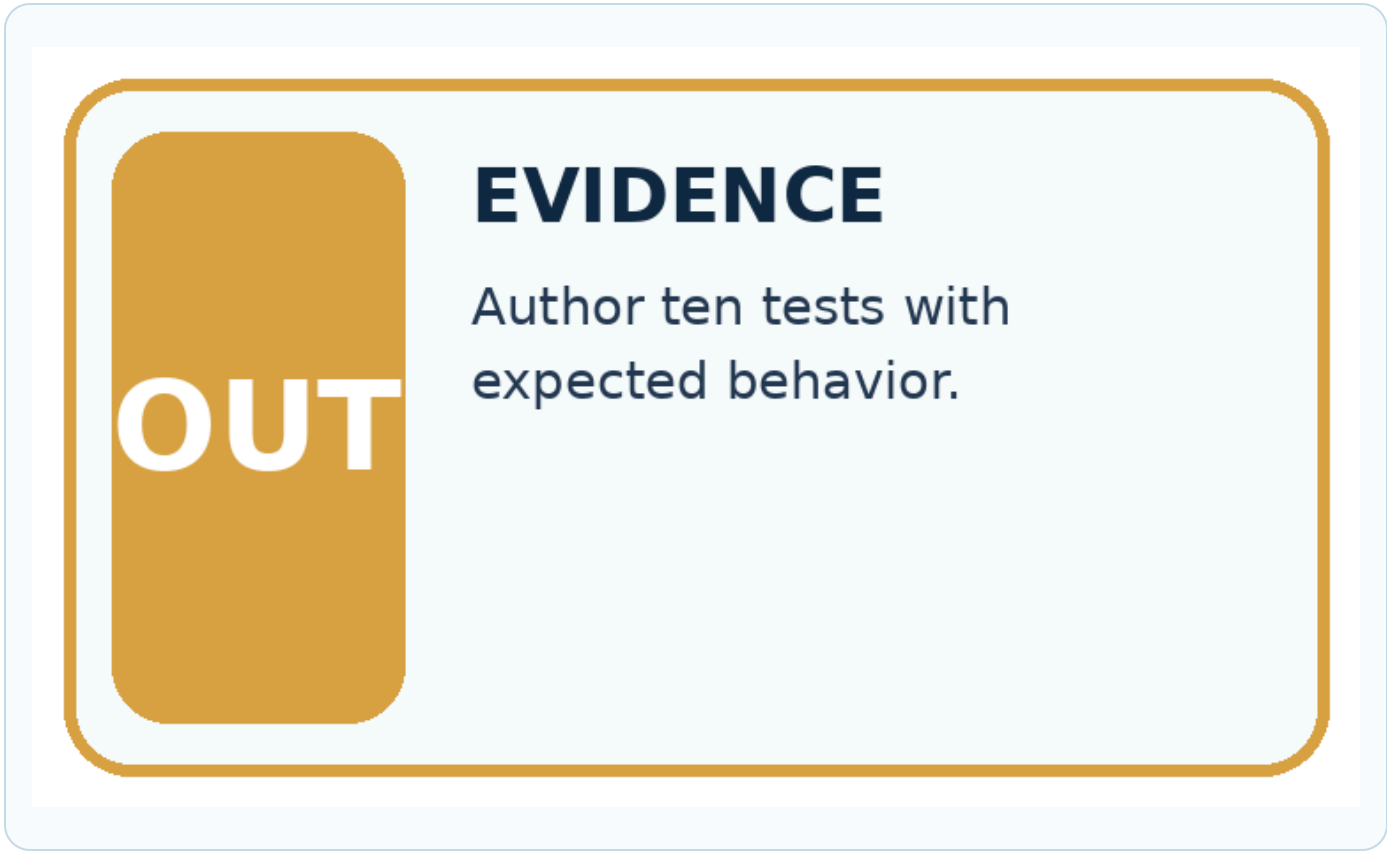
Trace the information. At each stage, ask what changed and how you could test it.


IN
INPUT
Author ten tests with expected behavior.

↓


SYSTEM
A good eval set represents real use, edge cases, and known risks.

↓



Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

Record case ID, **evidence** version, expected claims or fallback, permitted actions and scoring rubric. Separate development from final cases. Aggregate metrics only after checking required conditions individually.

System evaluation sets — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Ten cases include six answerable and four missing-evidence cases. System answers five answerable correctly and abstains appropriately on three missing cases.	Record case ID, evidence version, expected claims or fallback, permitted actions and scoring rubric. Separate development from final cases. Aggregate metrics only after checking required conditions individually.	Answer correctness 5/6; appropriate fallback 3/4; combined contract passes 8/10 if each case has one required outcome. Different denominators answer different questions.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

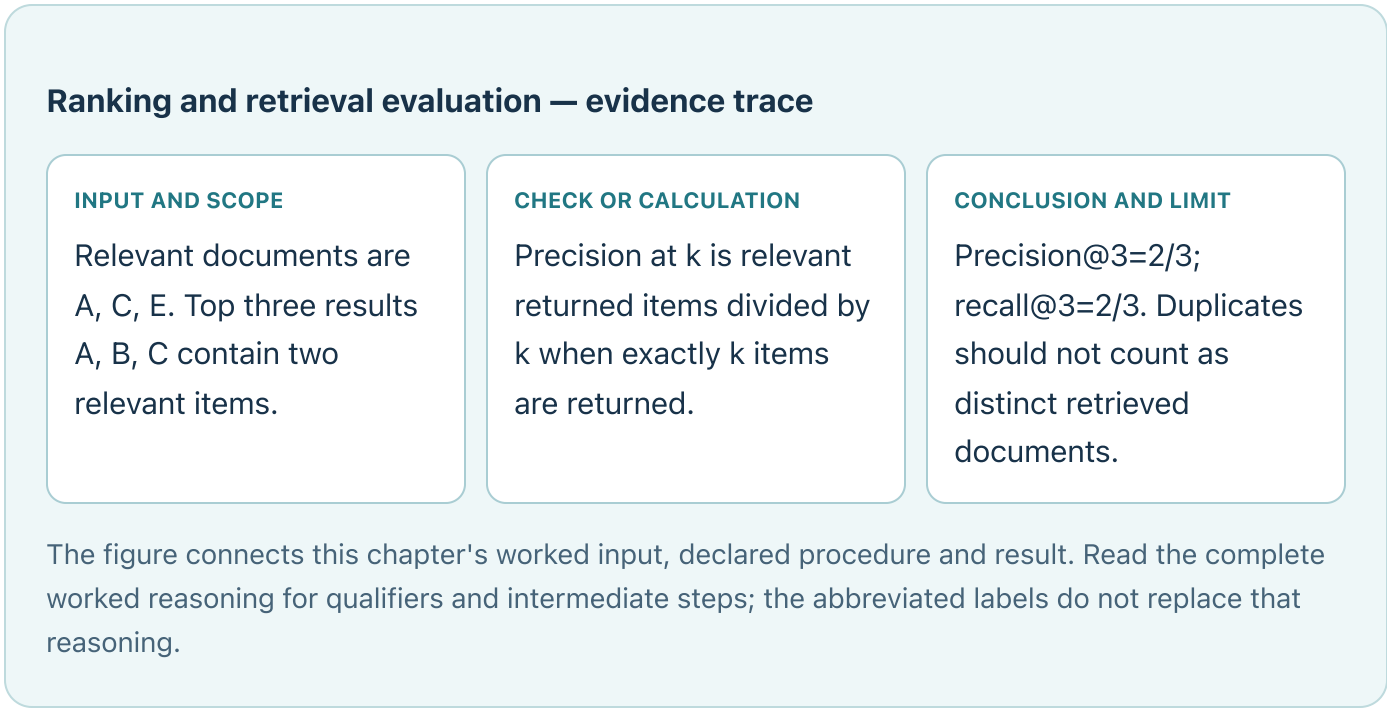
Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Apply the connected idea: Ranking and retrieval evaluation

Precision at k is relevant returned items divided by k when exactly k items are returned. **Recall at k** is relevant returned items divided by all known relevant eligible items.

If the relevant set is unknown, do not invent recall.



Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

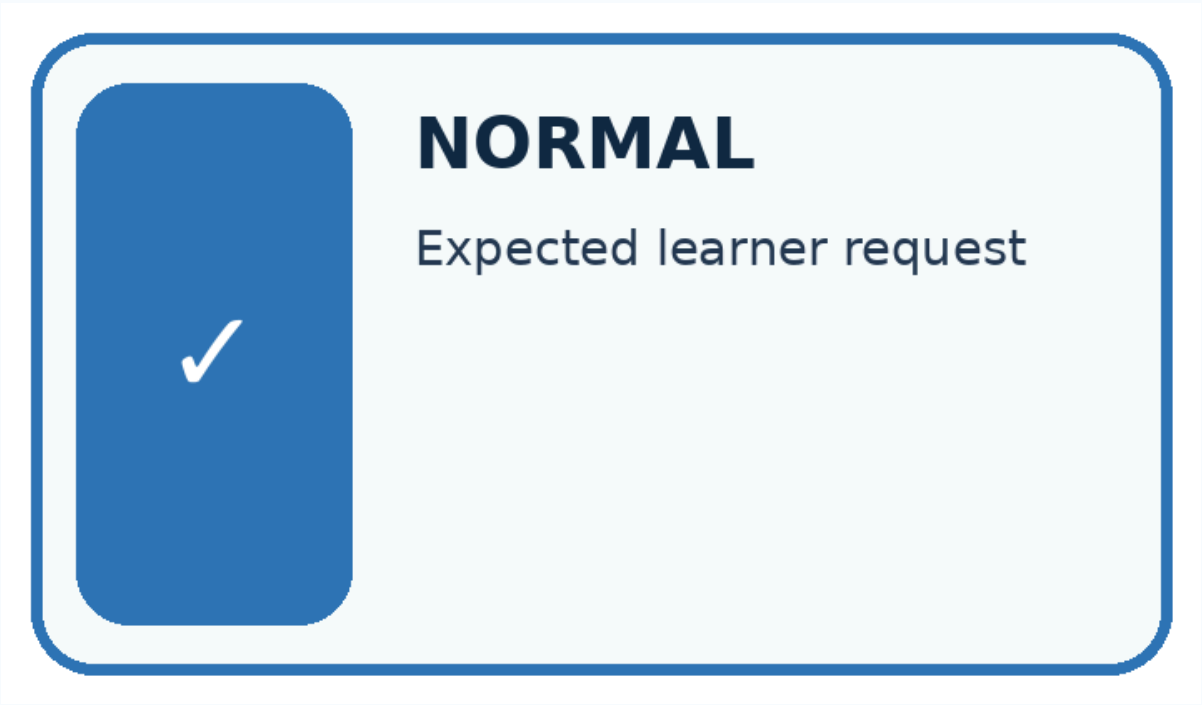
> Compare the supplied cases for Evaluation Sets. Identify the detail that changes the conclusion.

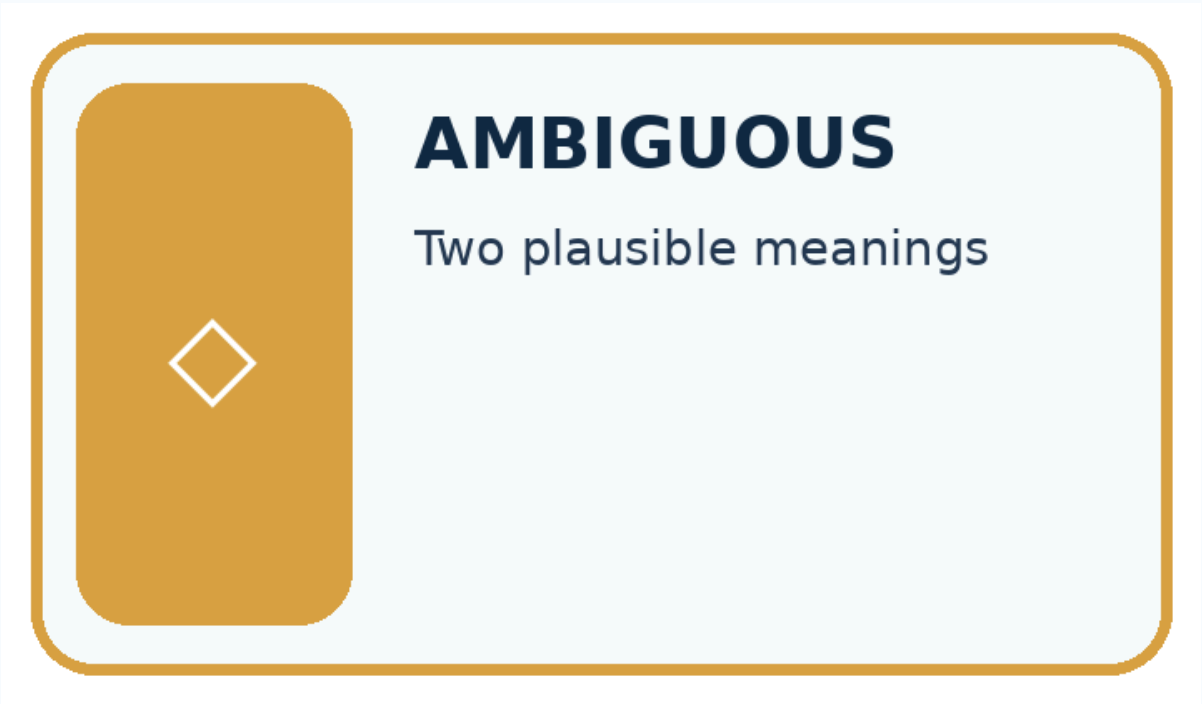
The case to investigate

CASE GALLERY

Four Tests, Four Kinds of **Evidence**

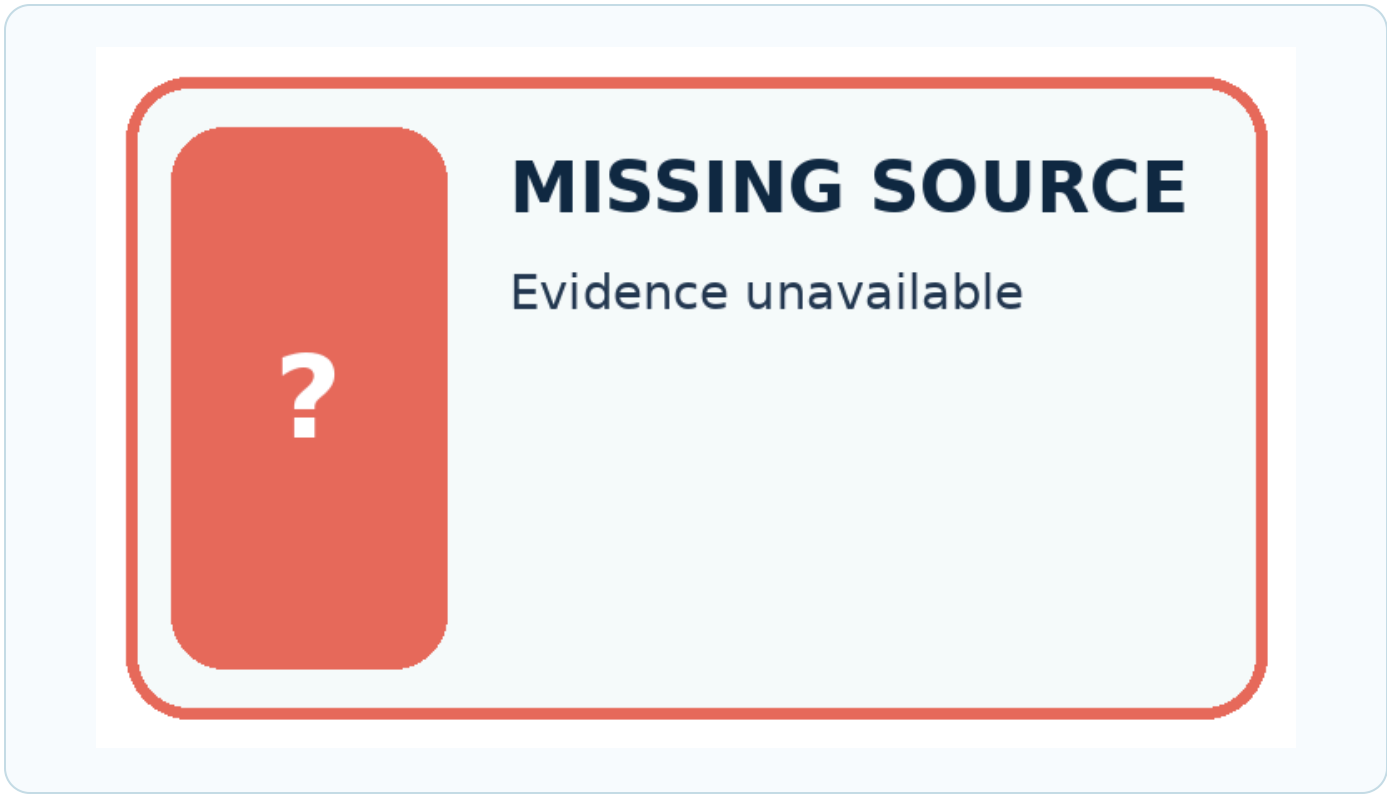
A system is not understood until it survives more than one comfortable example.

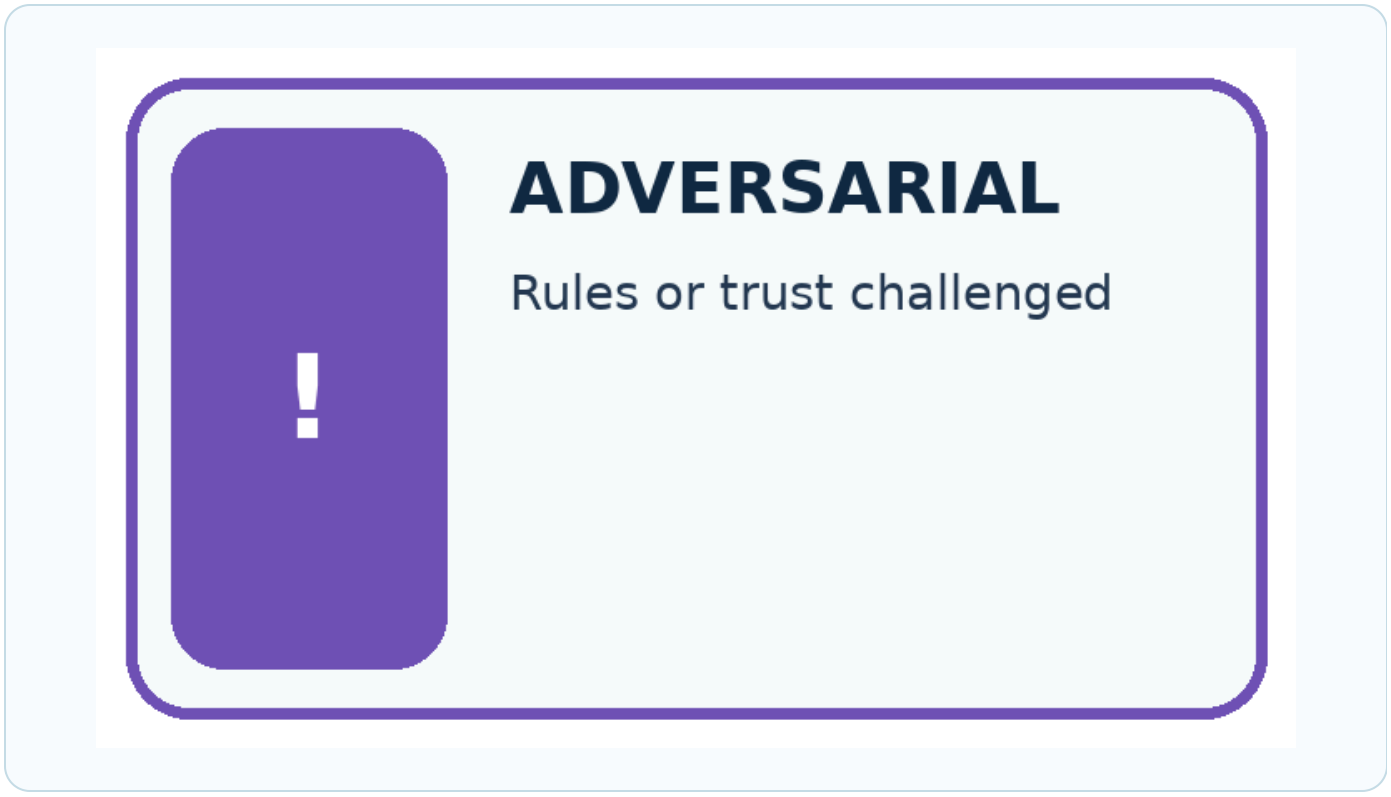




The normal case

NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.


?
MISSING SOURCE
Evidence unavailable


!
ADVERSARIAL
Rules or trust challenged

Check missing information

MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change one thing

Change: Add three missing-**evidence** cases that all pass.

Predicted result

Prediction: Overall becomes 8/9, but supported answering remains 2/3.

Why the result follows

Reason: Changing the case mixture raises the pooled rate without improving any existing answerable result.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

> Use the vocabulary of Evaluation Sets to explain a concrete decision, rather than only memorising definitions.

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.




2
COVERAGE
important situations included


3
EXPECTED BEHAVIO
observable acceptable response



Word	My own example	Not this / boundary
evaluation set	Cases used to measure a system under declared criteria; specify whether they are validation or final test cases.	
coverage	The range of relevant cases, conditions or groups included in data or tests.	
expected behavior	The result or action required by the specification for a given case, stated before observing the system.	
regression test	A repeatable check that previously correct behaviour still works after a change.	

Words used in the closer explanation

Word	Meaning
Evaluation case	An input with evidence and prespecified expected behaviour
Contract pass	Satisfying the declared requirements for one case

Terms for Ranking and retrieval evaluation

Term	Meaning
Precision at k	Relevant results among the first k returned items
Recall at k	Fraction of known relevant eligible items found in the first k results
Cutoff	The rank limit used for scoring

STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Evaluation Sets, predict a step, then compare with its explanation.

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

WEEK 26 • WORKED EVIDENCE

1. Ten cases include factual, comparison, calculation, ambiguous, and missing-source questions.
2. Add conflicting-source, prompt-injection, sensitive-data, and accessibility cases.
3. Write expected behavior before running the product.
4. After a repair, keep the failed case as a regression test.

1. Ten cases include factual, comparison, calculation, ambiguous, and missing-source questions.

2. Add conflicting-source, prompt-injection, sensitive-data, and accessibility cases.

3. Write **expected behavior** before running the product.

4. After a repair, keep the failed case as a **regression test**.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Ten cases include six answerable and four missing-**evidence** cases. System answers five answerable correctly and abstains appropriately on three missing cases.

Answer correctness 5/6; appropriate fallback 3/4; combined **contract pass**es 8/10 if each case has one required outcome. Different denominators answer different questions.

Worked explanation · Evaluation Sets

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Apply the connected idea: Ranking and retrieval evaluation

Relevant documents are A, C, E. Top three results A, B, C contain two relevant items.

Precision@3=2/3; recall@3=2/3. Duplicates should not count as distinct retrieved documents.

Worked explanation · Evaluation Sets

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Evaluation Sets

Design an **evaluation set** that checks answering and appropriate uncertainty separately.

The facts we will use

Supplied facts

Six toy questions: three have source answers and three do not. Results: 2 supported answers out of 3; 3 correct “not in source” responses out of 3.

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-3/26

Evaluation Sets

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Separate cases

Keep answerable and missing-evidence cases as different groups.

2. Calculate rates

Answering: 2/3. Appropriate uncertainty: 3/3.

3. Report total

Five of six pass overall; the perfect missing-evidence group hides one answering failure if only totals are discussed.

CHANGE ONE THING

Add three missing-evidence cases that all pass.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Separate cases	Keep answerable and missing-evidence cases as different groups.
2. Calculate rates	Answering: 2/3. Appropriate uncertainty: 3/3.
3. Report total	Five of six pass overall; the perfect missing-evidence group hides one answering failure if only totals are discussed.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Read the picture in order. Five of six pass overall; the perfect missing-**evidence** group hides one answering failure if only totals are discussed. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

Keep the stated limit

A small constructed set probes specific behaviours; its pooled rate is not an estimate of every future classroom question.

Keep the stated limit

Read the labelled picture: Did the added cases demonstrate better answering? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Keep the stated limit

Change one thing: Add three missing-**evidence** cases that all pass. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Evaluation Sets claim and explain a repair using the supplied evidence.

Find the first error

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: More random test cases always create better coverage. Strong move: inspect the claim, source, stage, schema, or test evidence; repair one cause; rerun.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Check the claim against the facts

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

Apply the connected idea: Ranking and retrieval evaluation

Check the claim against the facts

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

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STUDENT LESSON · 8 OF 18

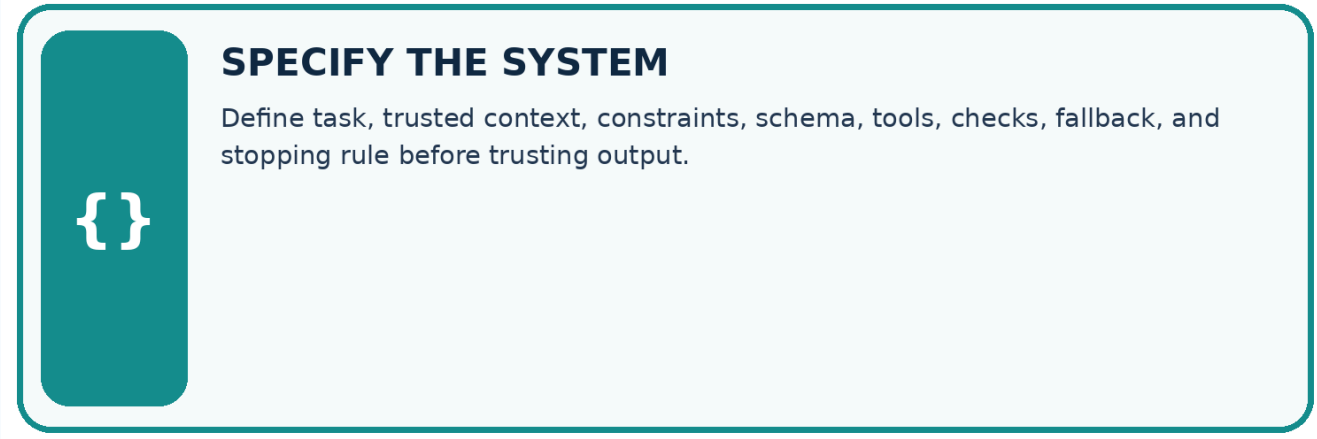
Practise together

> Practise Evaluation Sets with support, then explain the deciding step in your own words.

LOGIC STUDIO

Specify Before You Run

Prompts, schemas, retrieval, tool calls, and code are inspectable parts of one AI product.



SET task, audience, constraints, **output**_schema LOAD approved_context or retrieval_result CALL model or allowed_tool VALIDATE fields, claims, citations, and safety IF **evidence** missing: ask, retrieve, or abstain RETURN useful **output** + source trail + limitations

Trace one **input**. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

Evaluation **coverage** limits the claim; prompts tuned on these cases need new reserved final cases.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Apply the connected idea: Ranking and retrieval evaluation

Known judgments may be incomplete; a score over one query does not establish **coverage** over all queries.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

> **Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

Practise with less help: Trace the worked case

Use workbook W2 for the connected example “System **evaluation set**s”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for System **evaluation set**s. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Practise the connected topic: Ranking and retrieval evaluation

Use its named workbook W2. Identify the given **input**s, try the middle steps with less help, then explain your trace. Check this claim before group discussion: The worked result establishes every future case

Explain the picture

Did the added cases demonstrate better answering? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Evaluation Sets. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Map user tasks and risks, author ten labeled cases, peer-audit missing **coverage**, then freeze **expected behavior**s.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

> **Test / input:** boundary
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

> **Test / input:** missing
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

> **Test / input:** fresh transfer
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

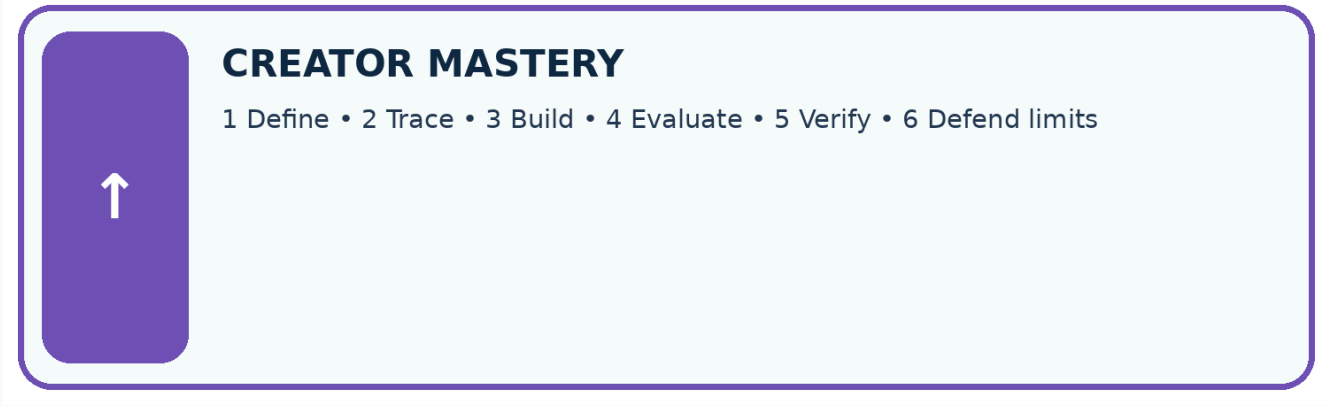
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Evaluation Sets. Explain why each step is justified.

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



1 • NAME: Define the concept without repeating the textbook.

2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.

3 • APPLY: Solve a different example independently.

4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest **evidence** of independent understanding:

Write or draw your reasoning and evidence.

STUDENT LESSON · 11 OF 18

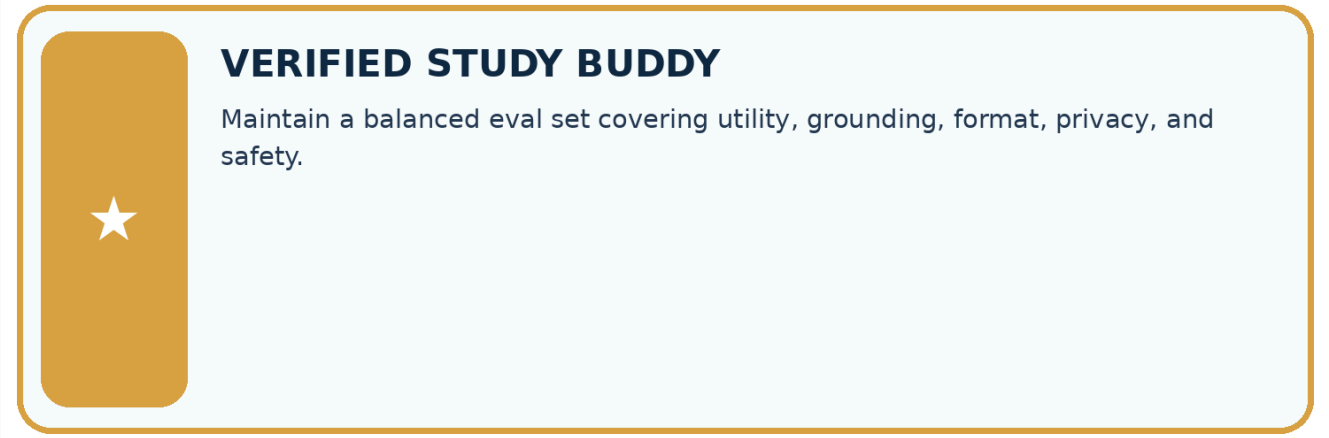
Connect your project

Apply this week's idea to Verified Study Buddy. Keep an annotated decision and the evidence supporting it.

CUMULATIVE PROJECT

Verified Study Buddy

Every week adds one inspectable piece. The system must show how it reached a result.



THIS WEEK'S CHECKPOINT Maintain a balanced eval set covering utility, grounding, format, privacy, and safety.

Trace the system

SYSTEM MAP Learner need → prompt/schema → approved sources/retrieval → model/tool → structured answer → verification → **human decision**

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under **human control**?

Write or draw your reasoning and evidence.

Responsible design

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse fluent **output** with verified truth.

Evidence for your project

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

STUDENT LESSON · 12 OF 18

Show what you know

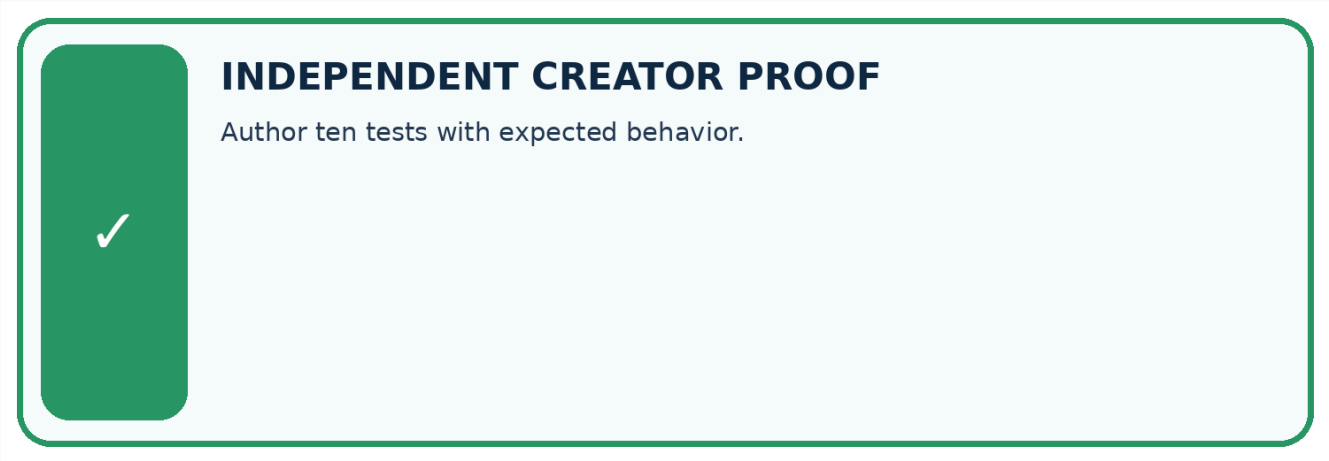
Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Independent evidence

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



Try a more demanding case

CHALLENGE Author ten tests with **expected behavior**.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My **evidence** that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

Check your explanation

SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Evaluation Sets case for supported practice. Repeated examples do not become fresh assessments.

Week 26 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

Ten evaluation questions are easy definitions; none lacks a source.

The question

> What is untested?

Worked explanation

Absent **evidence**, ambiguity and boundary handling. Add predefined fallback cases and keep final cases separate from tuning.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Empty evaluation categories reveal gaps

Evidence or stage	Value or result
Easy definitions	10

Evidence or stage	Value or result
Absent evidence	0
Ambiguity	0
Boundary handling	0

Ten evaluation questions are easy definitions; none lacks a source.

> What is untested?

Absent **evidence**, ambiguity and boundary handling. Add predefined fallback cases and keep final cases separate from tuning.

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Changed independent case

Approved notes: Robotics Tuesday Room2; Art Wednesday Room5. Requests: “Where does the club meet?” and “Who leads drama?” An unversioned added note says Robotics Room3. Design checks and safe responses.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 26 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

You improve a prompt after reading every test question. Can those questions still provide an untouched final evaluation? Propose a repair.

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Maintain a balanced eval set covering utility, grounding, format, privacy, and safety.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

A test set contains ten easy questions with direct answers in the notes. Add three different kinds of cases that probe fallback behaviour and define expected outcomes.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Evaluation Sets, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Design an **evaluation set** that checks answering and appropriate uncertainty separately.

By the end, I can	How I will show it
Explain the mechanism and terminology of system evaluation sets	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Evaluation Sets without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of system **evaluation set**s

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Design an **evaluation set** that checks answering and appropriate uncertainty separately.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Five of six pass overall; the perfect missing-**evidence** group hides one answering failure if only totals are discussed.

1. Separate cases

2. Calculate rates

3. Report total

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Report total Five of six pass overall; the perfect missing-**evidence** group hides one answering failure if only totals are discussed. Read the picture in order.

Five of six pass overall; the perfect missing-**evidence** group hides one answering failure if only totals are discussed. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Add three missing-**evidence** cases that all pass. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. Overall becomes 8/9, but supported answering remains 2/3.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Overall becomes 8/9, but supported answering remains 2/3. Changing the case mixture raises the pooled rate without improving any existing answerable result.

Repair a missing building block

Review Quality Rubrics (ai-3/25). Goal: Explain the **mechanism** and terminology of system **evaluation set**s. Short practice: Why is “one out of two checks passed” an insufficient release decision?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Block to Python Bridge (ai-3/23). Goal: Translate a short paper calculation into a Python script. Short practice: Trace the changed list and identify where the result differs.

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Conversation State and Memory (ai-3/20). Goal: Explain the **mechanism** and terminology of conversation state and retention. Short practice: What is selected_book after the correction and after exit?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

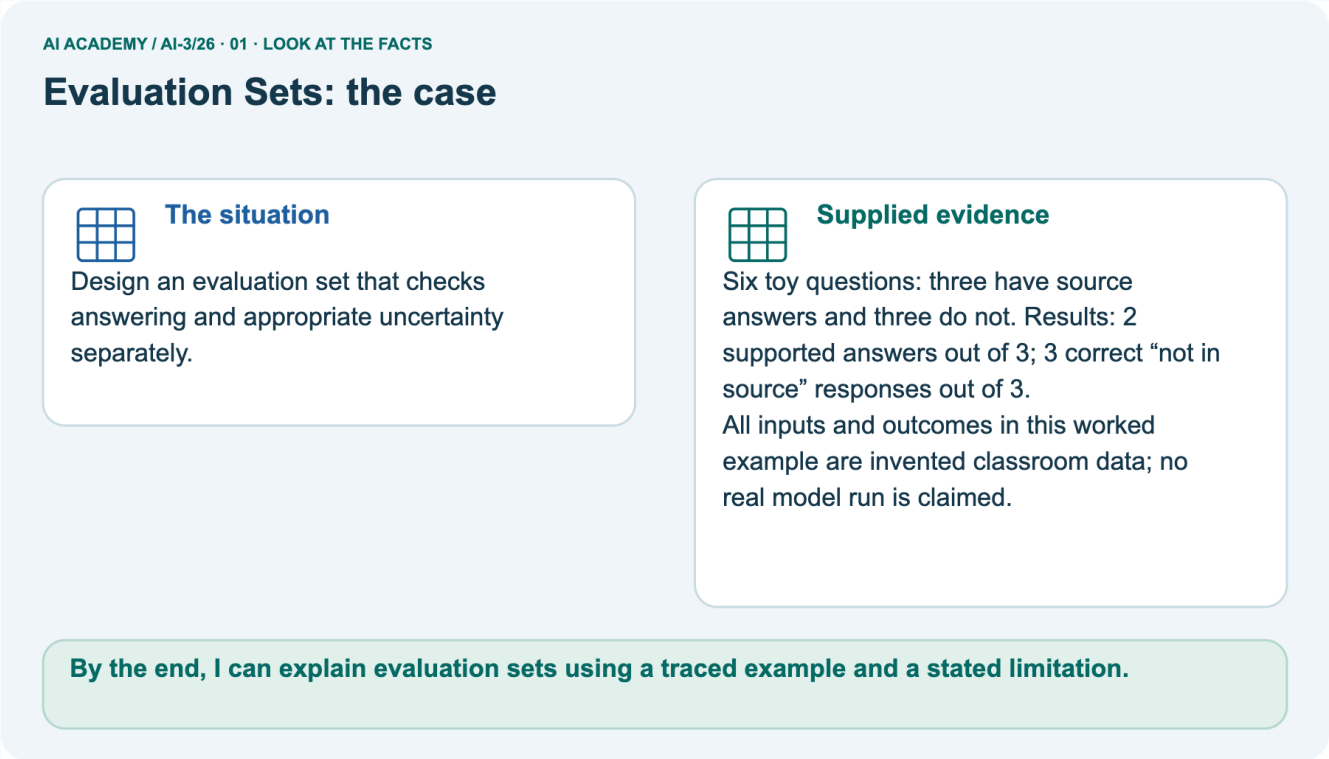
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Evaluation Sets graphic using its labels, deciding evidence and relevant limitation.

01. Evaluation Sets: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

Design an **evaluation set** that checks answering and appropriate uncertainty separately. Supplied facts: Six toy questions: three have source answers and three do not.

Results: 2 supported answers out of 3; 3 correct “not in source” responses out of 3. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



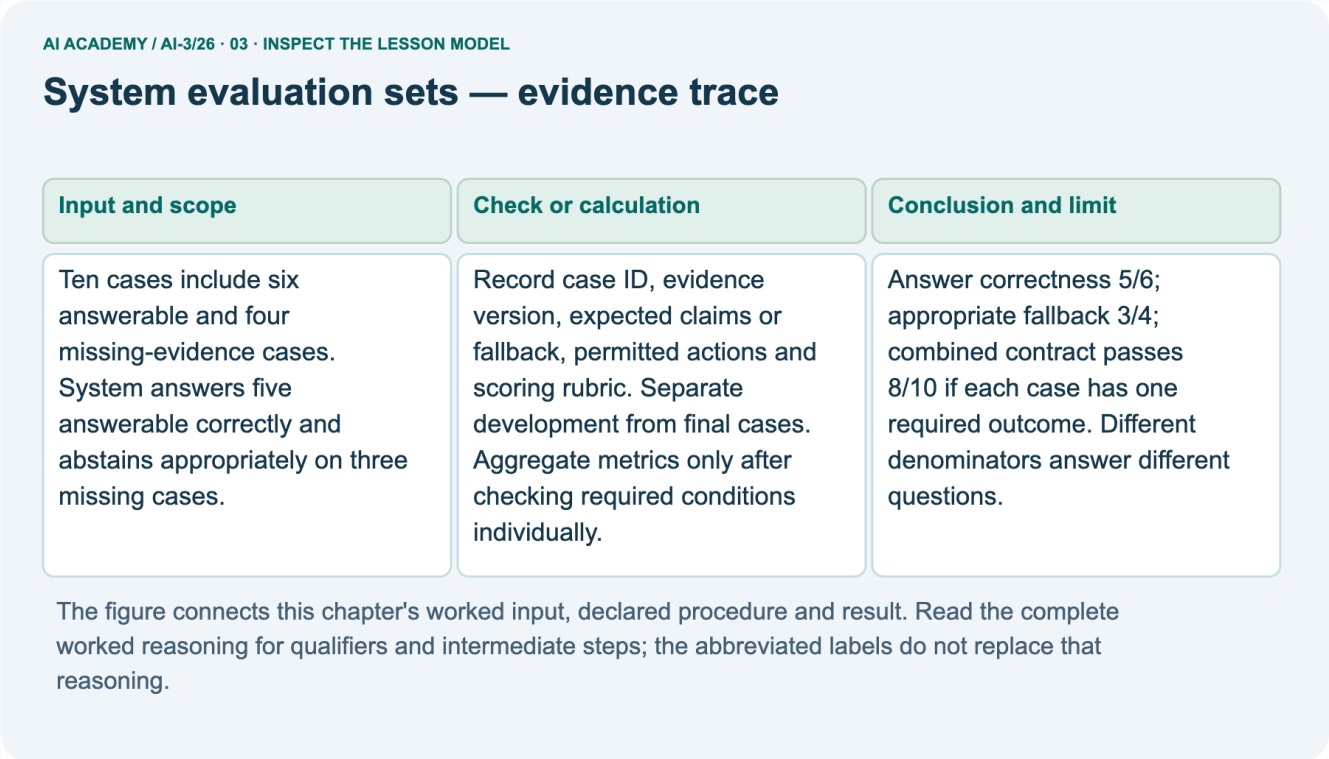
Follow the labelled picture with the matching explanation. Open the vector diagram

Separate cases: Keep answerable and missing-**evidence** cases as different groups. → Calculate rates: Answering: 2/3.

Appropriate uncertainty: 3/3. → Report total: Five of six pass overall; the perfect missing-**evidence** group hides one answering failure if only totals are discussed.

> Point to each numbered stage and explain which recorded **input** it uses.

03. System evaluation sets — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

System **evaluation set**s — **evidence** trace. Ten cases include six answerable and four missing-**evidence** cases.

System answers five answerable correctly and abstains appropriately on three missing cases. | Record case ID, **evidence** version, expected claims or fallback, permitted actions and scoring rubric.

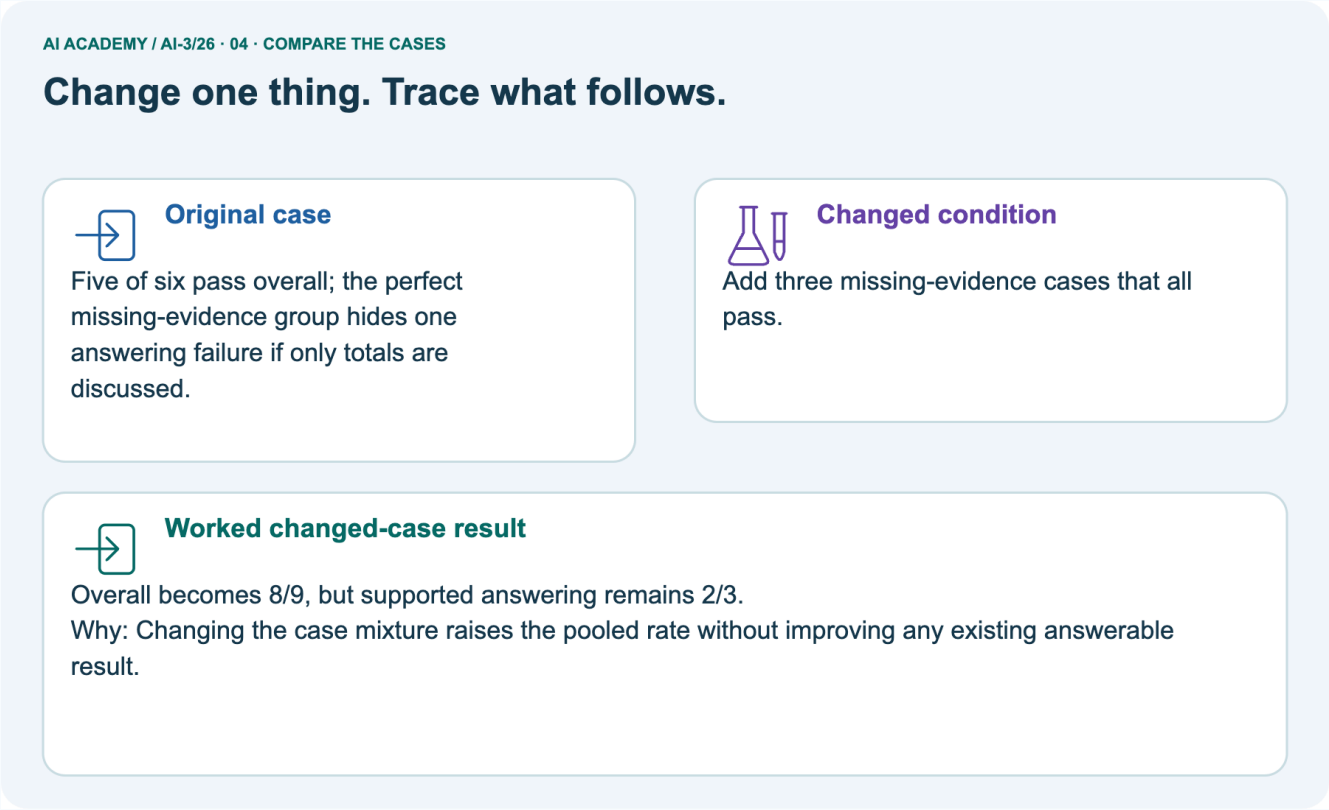
Separate development from final cases.

Aggregate metrics only after checking required conditions individually. | Answer correctness 5/6; appropriate fallback 3/4; combined **contract pass**es 8/10 if each case has one required outcome.

Different denominators answer different questions.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



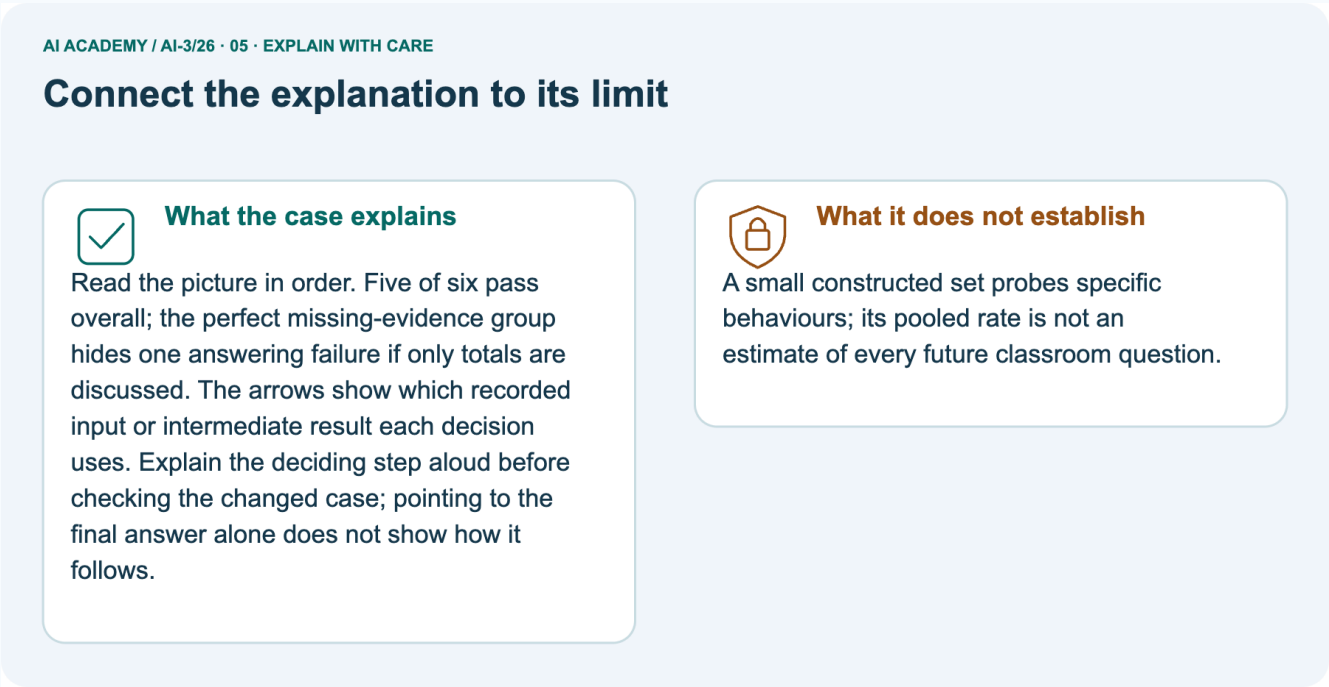
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Five of six pass overall; the perfect missing-**evidence** group hides one answering failure if only totals are discussed. Changed condition: Add three missing-**evidence** cases that all pass.

Expected result: Overall becomes 8/9, but supported answering remains 2/3. Reason: Changing the case mixture raises the pooled rate without improving any existing answerable result.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

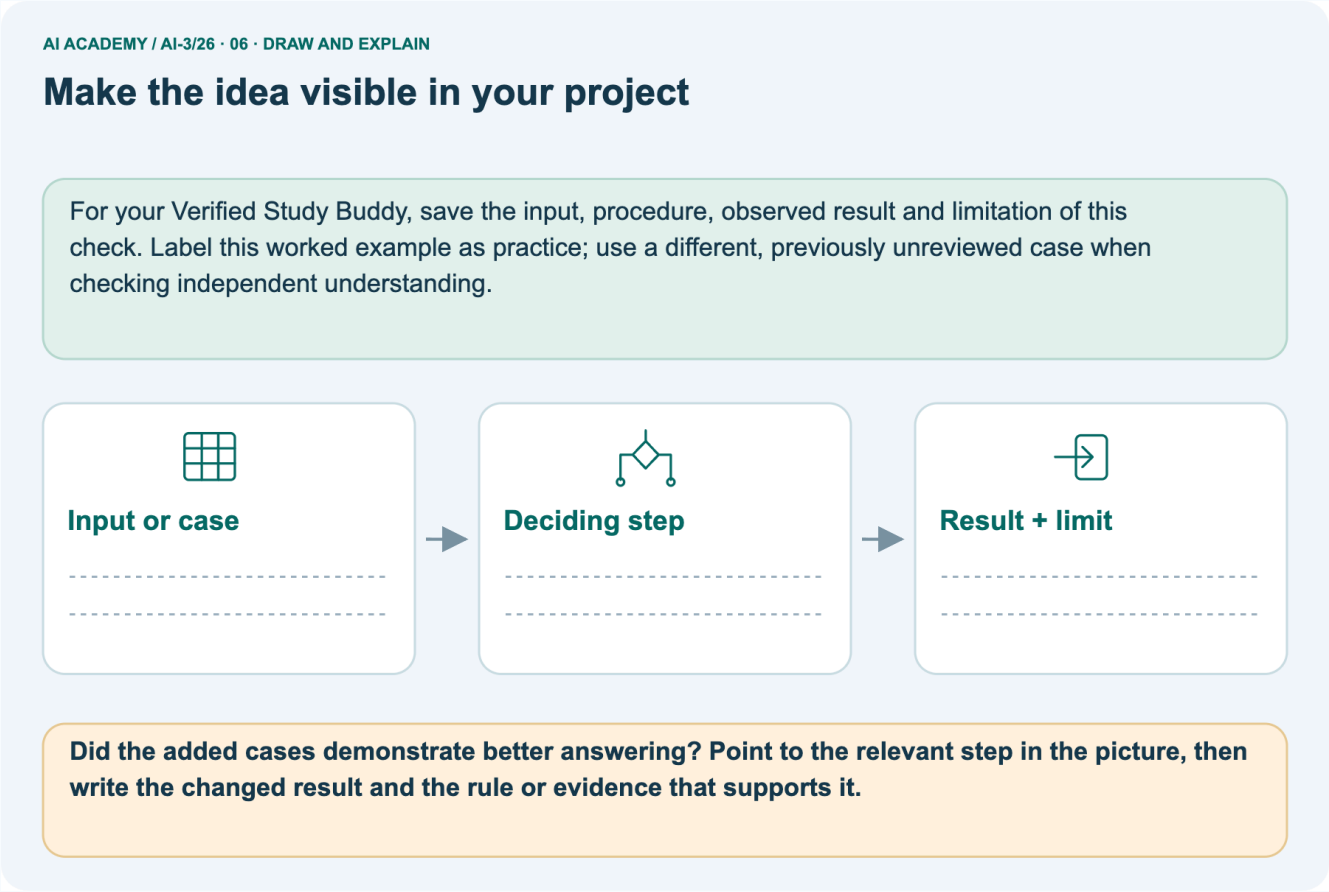
Read the picture in order. Five of six pass overall; the perfect missing-**evidence** group hides one answering failure if only totals are discussed. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: A small constructed set probes specific behaviours; its pooled rate is not an estimate of every future classroom question.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Did the added cases demonstrate better answering? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.